

Mobile Design Pattern

Course Teacher: Md. Mushfiqul Haque
Courtesy: Md. Atiqur Rahman

Navigation

The most common complaints about an application seem to revolve around:

- Crashing.
- Lack of key features (e.g., syncing, filtering, account linking).
- Confusing interface design.
- Poor navigation (e.g., can't go back, can't find things).

The first two issues can't be fixed with design patterns—they'll both require user and device testing—but the third and fourth complaints certainly can.

Following the common design patterns for navigation will ensure that people can find and use the valuable features in your application.

Primary Navigation Patterns

- Primary Navigation Pattern is used for primary navigation like navigating from one primary category to another.
- **Example:** top-level menus of a desktop application.

Primary navigation has evolved into two distinct types: **persistent and transient.**

- **Persistent navigation** encompasses simple menu structures like the List Menu and Tab Menu. As soon as you open an app with persistent navigation, it is immediately clear what the primary navigation options are.
- **Transient navigation**, however, must be explicitly revealed with a tap or gesture. **These patterns arise from the constraints of smartphone screen sizes, which have pushed mobile designers to think “outside of the box,” literally.**

Transient Navigation example using Ubuntu touch open mobile OS

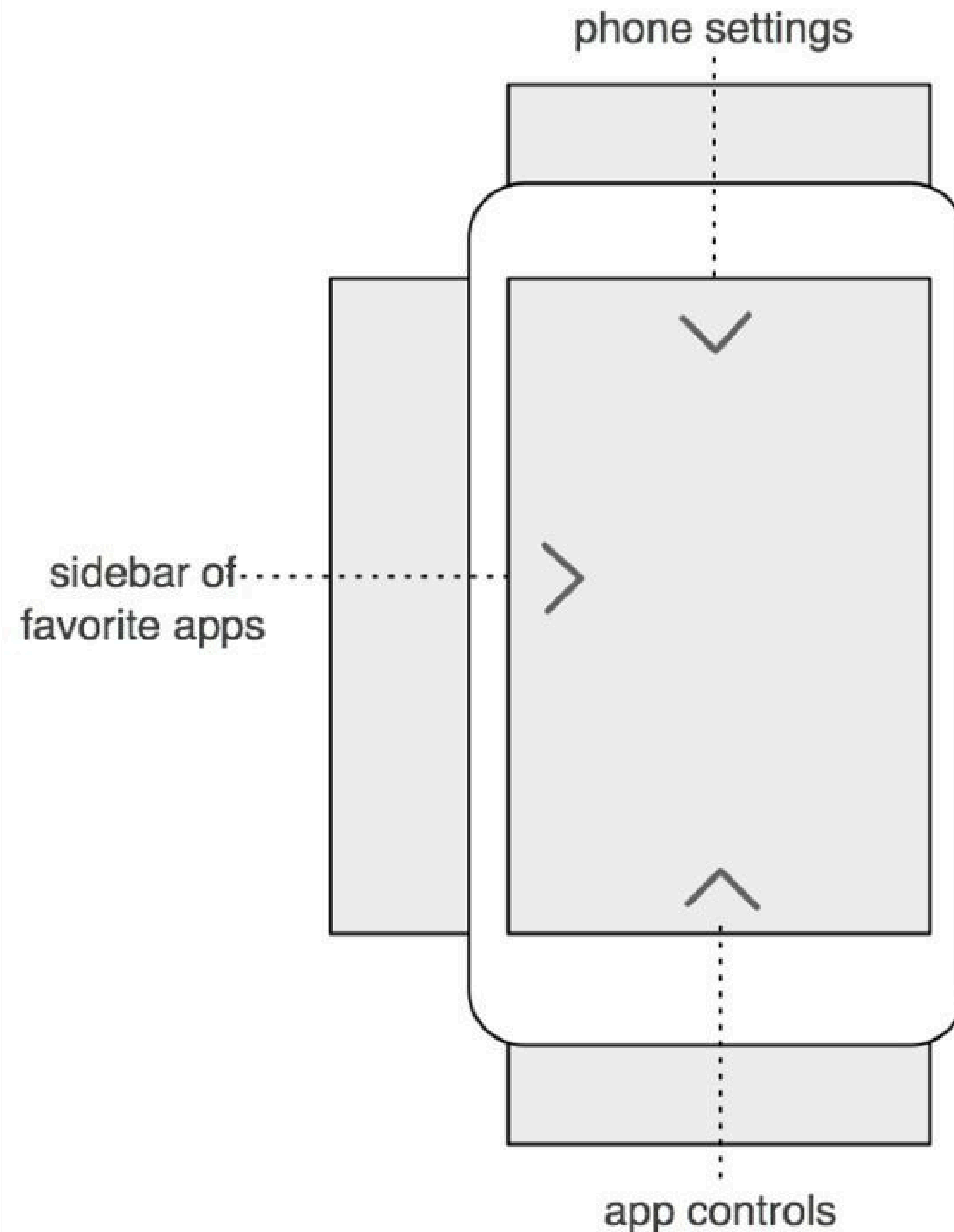


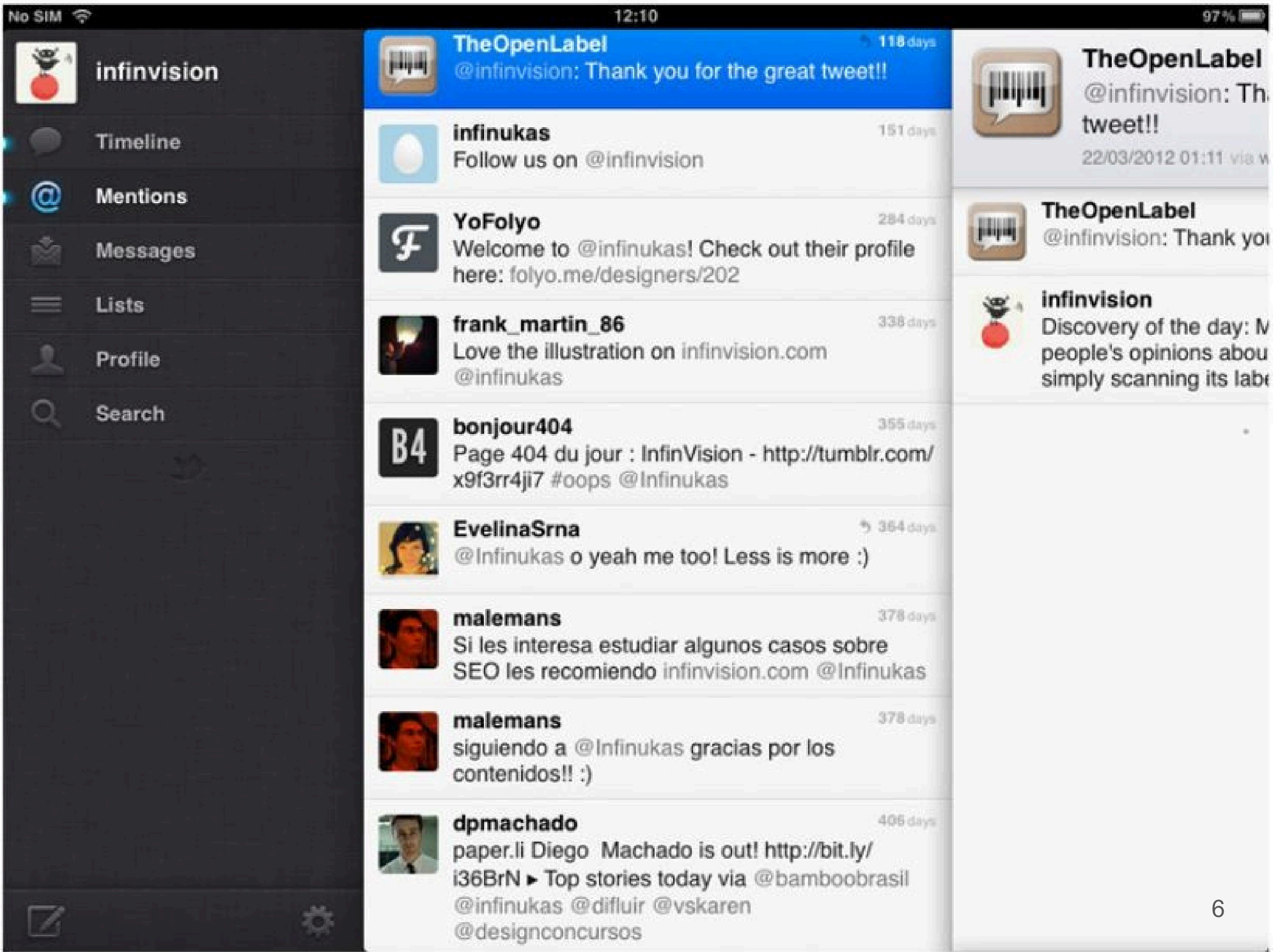
FIGURE 1-5.

In Ubuntu, you can swipe the screen edges to reveal settings and menus, leaving the screen entirely free for the application's content

3D layers and gestures

Twitter's early iPad design was a fantastic example of how 3D layers and gestures can create a uniquely mobile experience: the left panel is the menu bar, the middle panel is the listing of contents, and the right panel displays those contents. Tapping an item in the middle section collapses the left menu bar and shows a preview of the contents within the right panel. When tapped, the right panel expands to cover about 70% of the screen.

Example



Continued

No SIM 12:11 97%

TheOpenLabel
@infinvision: Thank you for the great tweet!! 118 days

infinukas
Follow us on @infinvision 151 days

YoFolyo
Welcome to @infinukas! Check out their profile here: folyo.me/designers/202 284 days

frank_martin_86
Love the illustration on infinvision.com @infinukas 338 days

B4
Page 404 du jour : InfinVision - http://tumblr.com/x9f3rr4ji7 #oops @Infinukas 355 days

EvelinaSrna
@Infinukas o yeah me too! Less is more :) 364 days

malemans
Si les interesa estudiar algunos casos sobre SEO les recomiendo infinvision.com @Infinukas 378 days

malemans
siguiendo a @Infinukas gracias por los contenidos!! :) 378 days

dpmachado
paper.li Diego Machado is out! http://bit.ly/i36BrN ▶ Top stories today via @bamboobrasil @infinukas @diflur @vskaren @designconcursos 406 days

bonjour404
Page 404 du jour : InfinVision - http://tumblr.com/x9f3rr4ji7 #oops @Infinukas 29/07/2011 10:01 via Tumblr

xoKee_Yan_Uh
RT "@inDependent_ox: #FREE @SHI_millz & @SwagChampNoodle ... they cant hold you love birds forever ... #oops ! BOL !." 1 hour

ohshannon3
Lol totally just sent a cute text saying I love you baby to my coworker Marta... #oops you're not Cam. Awk 1 hour

brittbirrrd
why do you act so snooty? you're not even pretty enough to do that. or pretty at all. #oops #sorrynotsorry 2 hours

emeriearellano
It felt like the right thing to say at that moment.. but, it really wasn't. #oops 3 hours

megankayglenn
Just skinny dipped with a minor... @MadisonMcDonnel #oops #mexicanprisonsarefunright 3 hours

jessiedarlingx
Waiting for a text then realizing you're the one 4 hours

7
Save Search

Continued

No SIM12:2695%

James @itsa

History Tusca orig...

pierotucci
History always fascinates me, Love and War in Tuscany (part 1)
trustandtravel.com/blog/iris-orig... via @twitter_username
1 day ago via Tweet Button

ENGLISH | DEUTSCH | FRANÇAIS | ABOUT | SUBSCRIBE TO KATHARINA'S ITALY

FOUNDER OF TRUST & TRAVEL

BROWSE BY REGION

All

Art & Culture

Food & Wine

Things to do

Fun with kids

Life at the villa

Shopping

Interior design

Love and War in Tuscany (part 1)

Art & Culture, Life at the villa, Tuscany

5

2

La Foce and Iris Origo - Love and War in Tuscany

How to decide between persistent and transient navigation

- Is your application “flat”? Are the menu categories equivalent in hierarchy, and are there just a few primary categories (i.e., three to five) in the app?
- Do your users need the menu to be always visible for quick access?
- Do the menu categories have status indicators, like the number of unread emails, for instance?

If you answered “yes” to one or more of these questions, it’s probably best to stick with persistent navigation. Now let’s take a look at those patterns.

SpringBoard

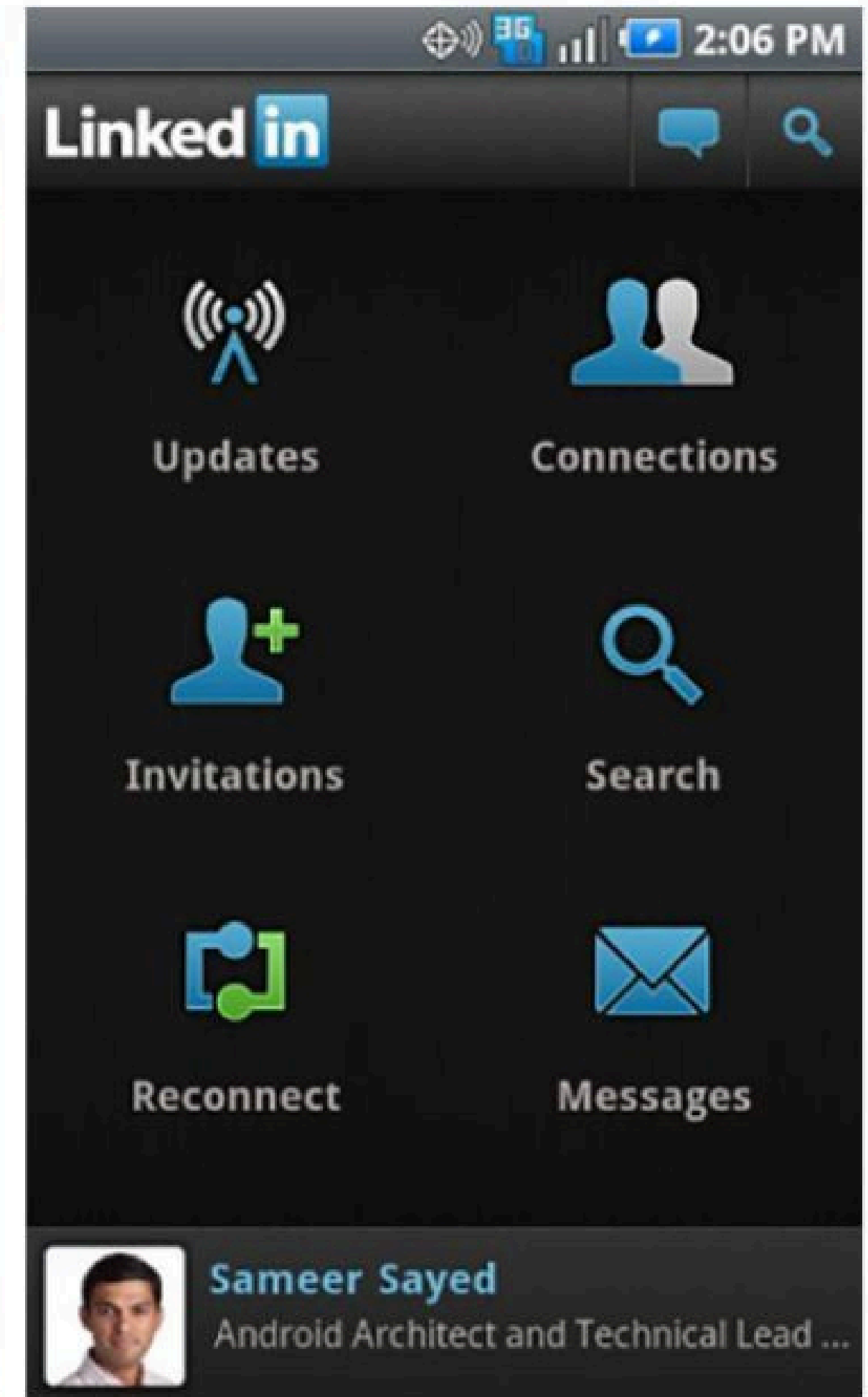
The Springboard pattern, also called a Launchpad, was the most popular navigation pattern in 2011. **This design is a landing screen with options that act as launch points into the application.**

It was also popular because up to nine options (in a 3×3 grid) could be displayed, compared to the limits of three to five tabs imposed by iOS and Android tab bars. And by adding a paging indicator (those little dots at the bottom), designers could provide even more menu options.

The main drawback of the Springboard pattern is that it flattens all options to the same level of importance.

FIGURE 1-8.

Facebook for iOS and
LinkedIn for Android:
Springboard designs
from 2011



Is SpringBoard Dead?

No, Android, iOS, and Windows Phone all still use this navigation pattern at the OS level.

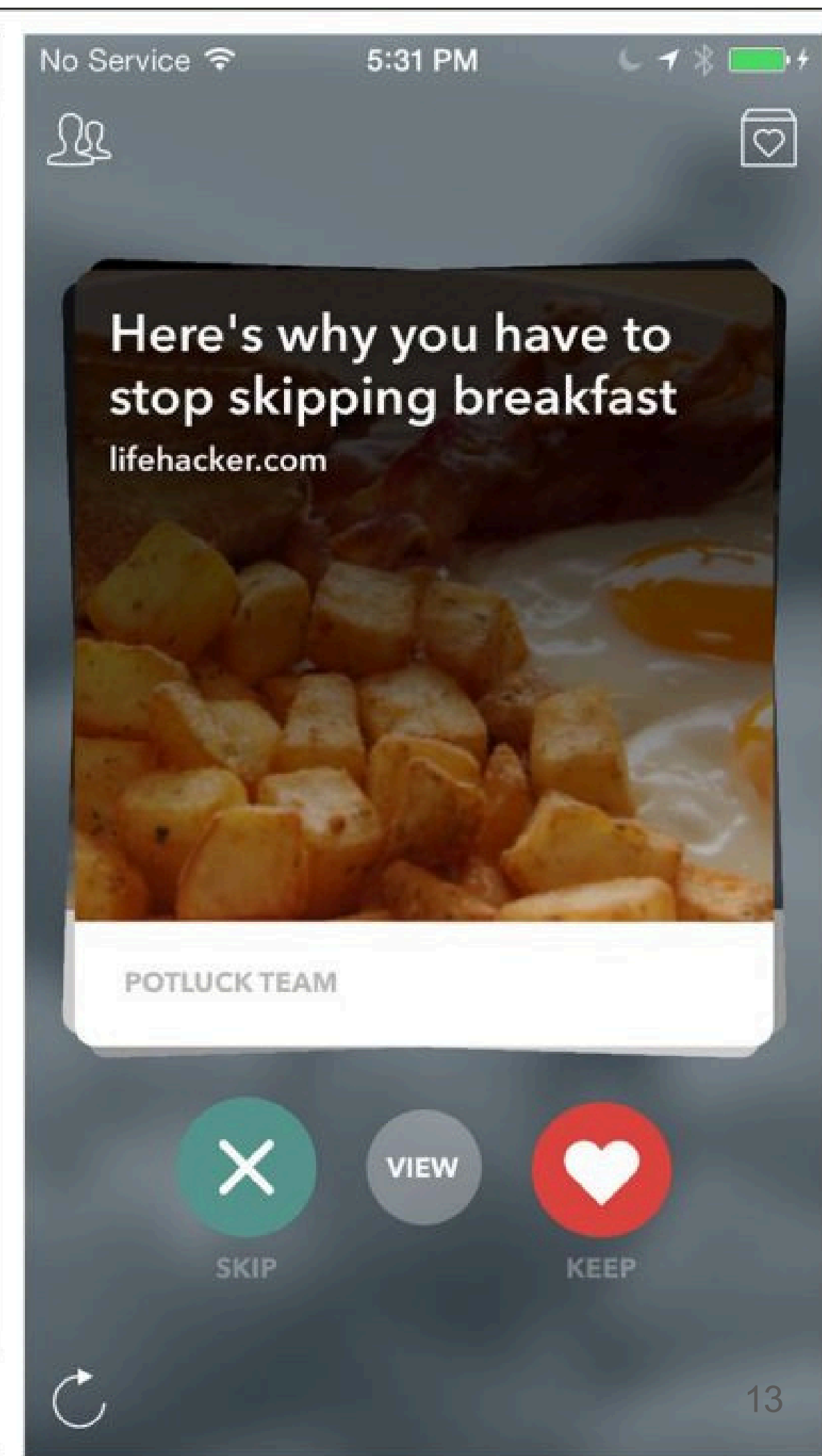
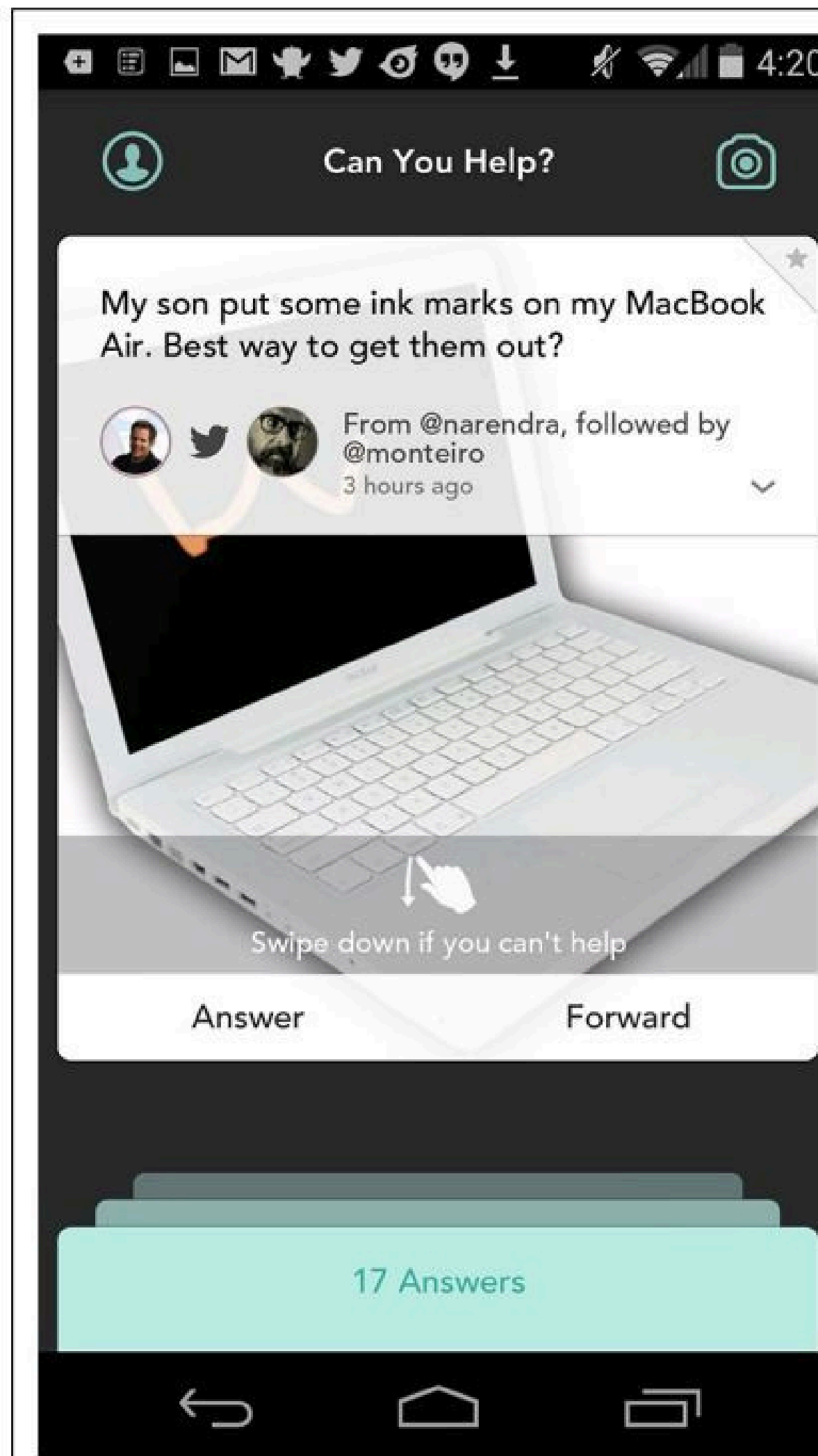


FIGURE 1-10.

iOS 7, Android KitKat, and Windows Phone 8 all use the Springboard pattern at the OS level

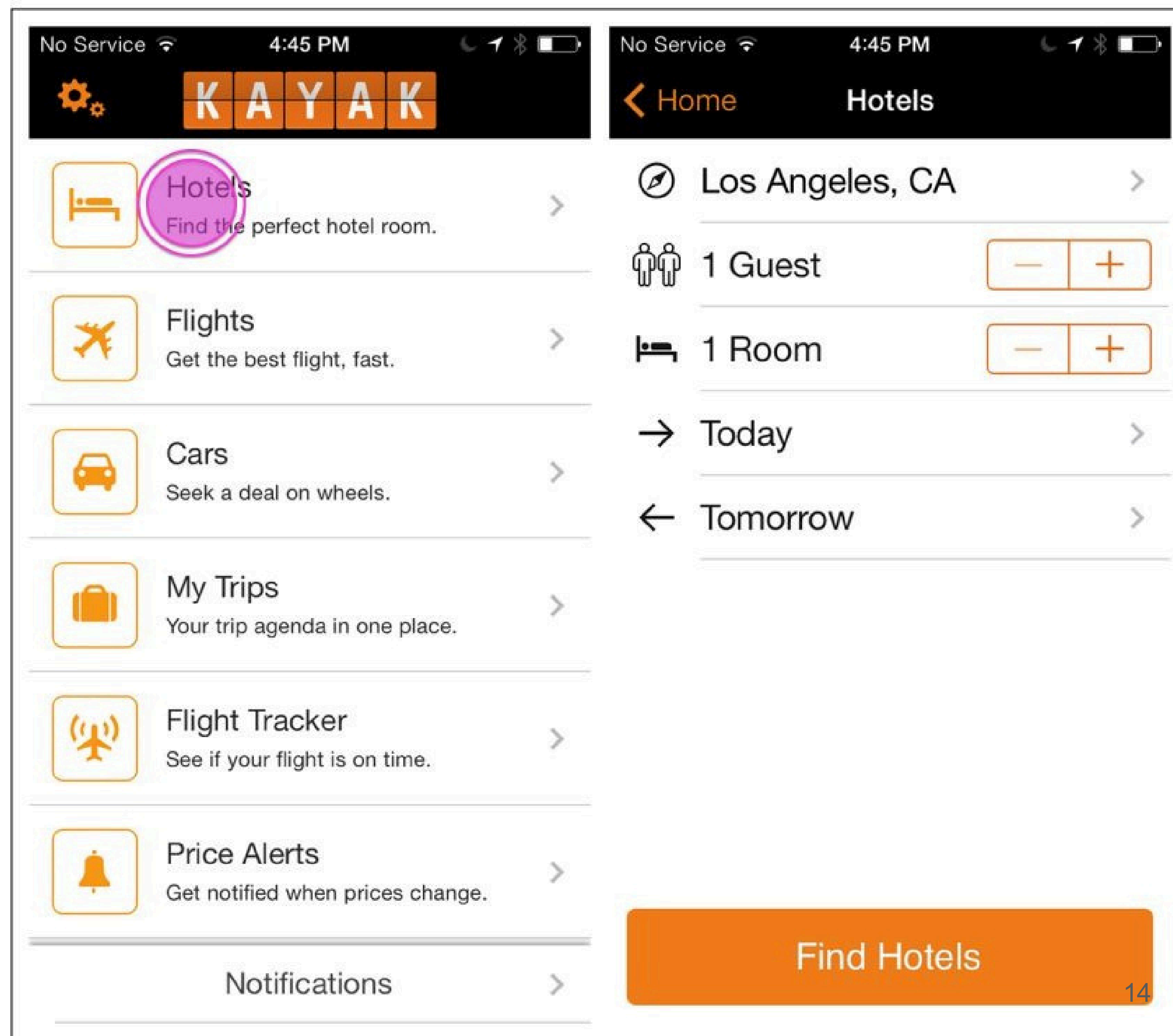
Cards

Card navigation is based on a card deck metaphor, including common card deck manipulations such as stacking, shuffling, discarding, and flipping.



List Menu

Similar to the Springboard in that each list item is a launch point into the application, and switching modules requires navigating back to the list. Apple calls this **hierarchical navigation**: where users navigate by making one choice per screen until they reach their destination. To navigate to another destination, users must retrace some of their steps—or start over from the beginning—and make different choices. Settings and Mail are good examples of apps that use a hierarchical structure.

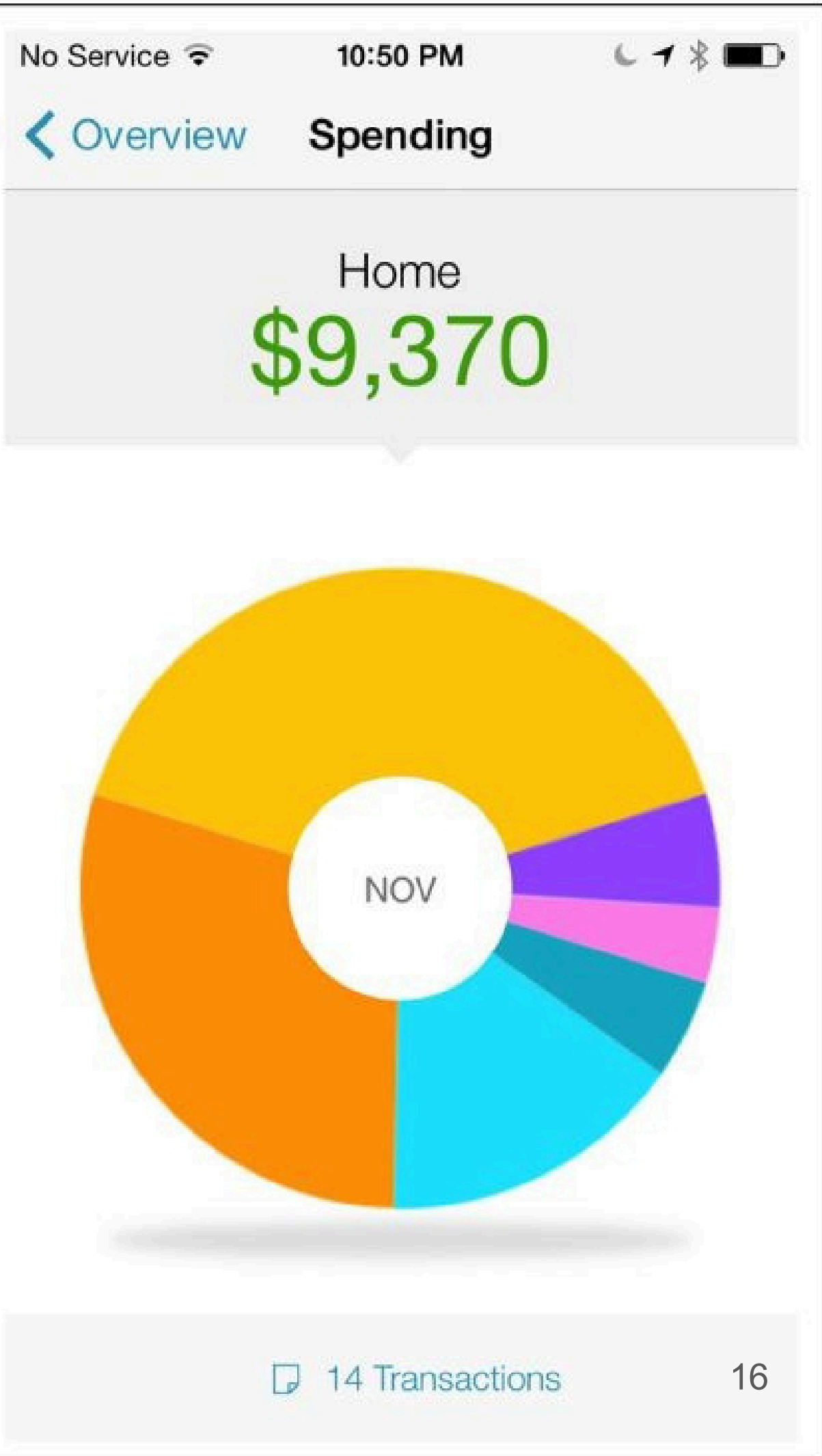


Up Button in List Menu

The Up button is used to navigate within an app based on the hierarchical relationships between screens. For instance, if screen A displays a list of items, and selecting an item leads to screen B (which presents that item in more detail), then screen B should offer an Up button that returns to screen A. If a screen is the topmost one in an app (that is, the app's home), it should not present an Up button.

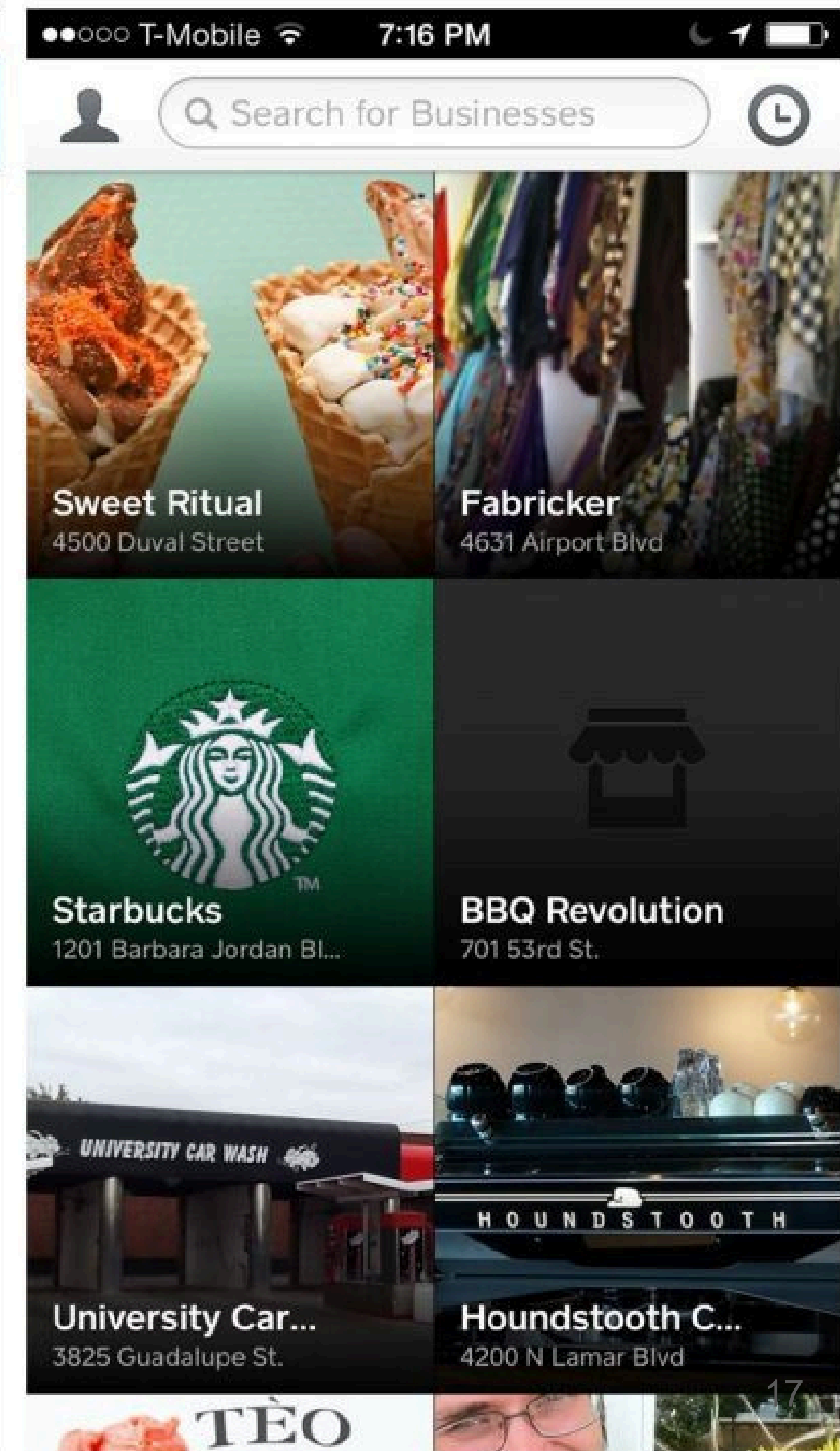
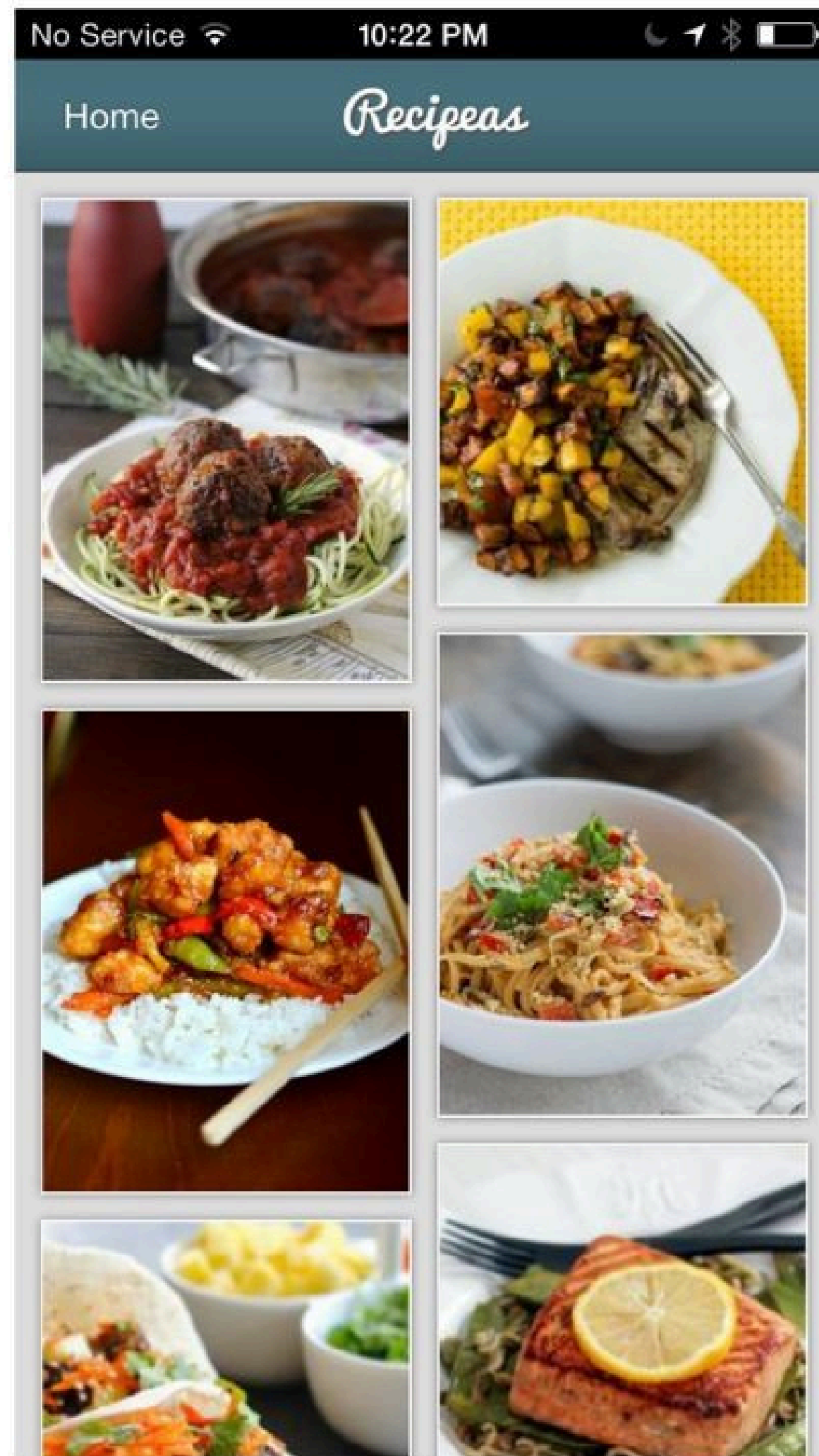
Dashboard

The Dashboard pattern is similar to the Springboard and List Menu patterns. With a quick glance, a good Dashboard gives the user a snapshot of the most relevant information he needs to know, without making him navigate into another screen.



Gallery

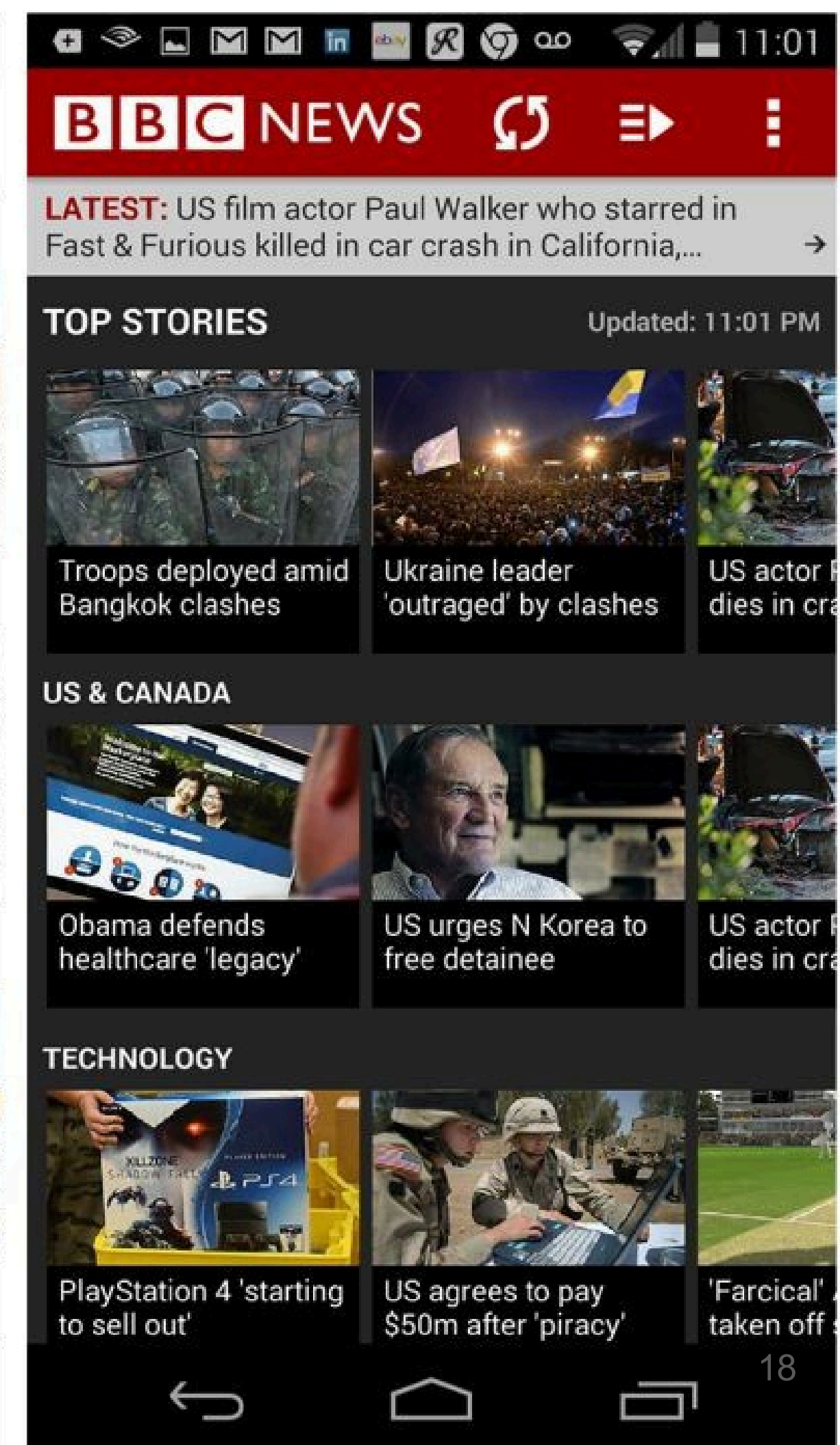
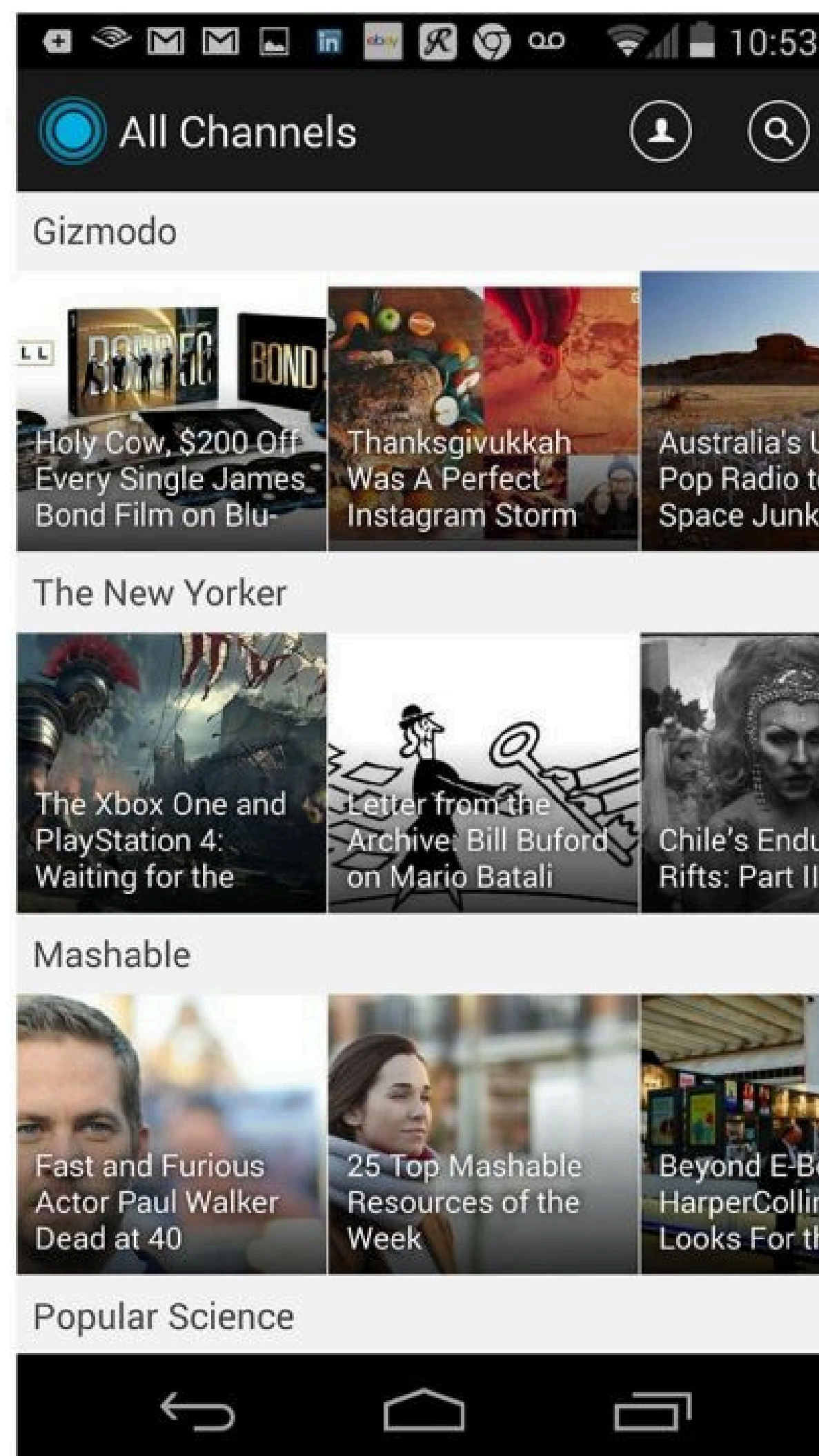
The Gallery pattern displays live content—like news stories, recipes, or photos—arranged in a grid, a carousel, or a slideshow.



Continued

Notice the BBC News example is easier to scan than the LinkedIn Pulse example, because the titles are below the photo instead of overlaid on them.

The Gallery pattern works best for showing frequently updated, highly visual content where no hierarchy is implied.



Tab Menu

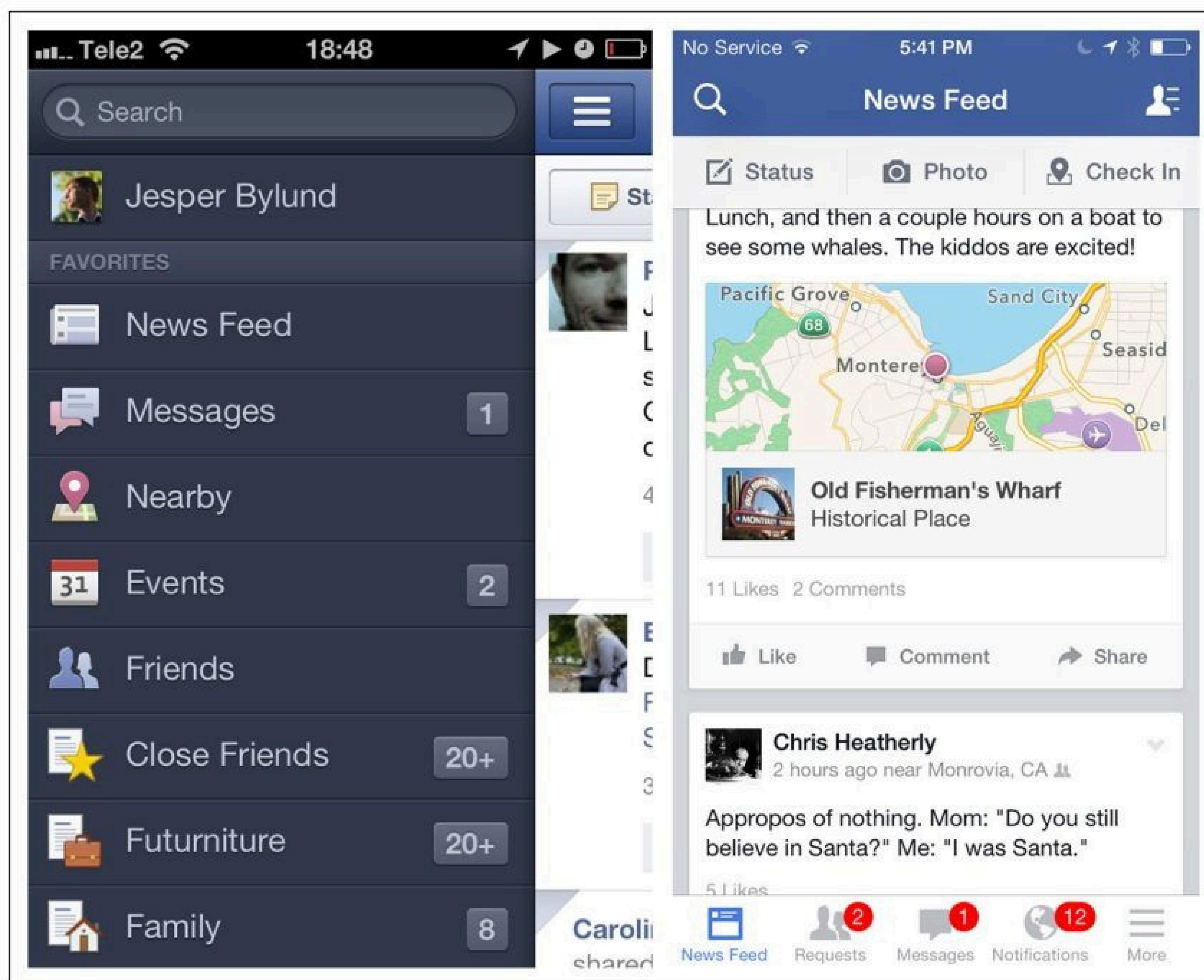
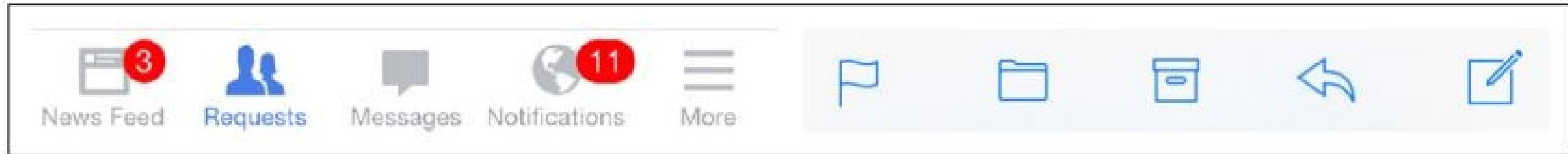


FIGURE 1-25.

Facebook for iOS, old and new: Tab Bar (right) beat out the Side Drawer (left) and other navigation patterns in 10-million-user test batches

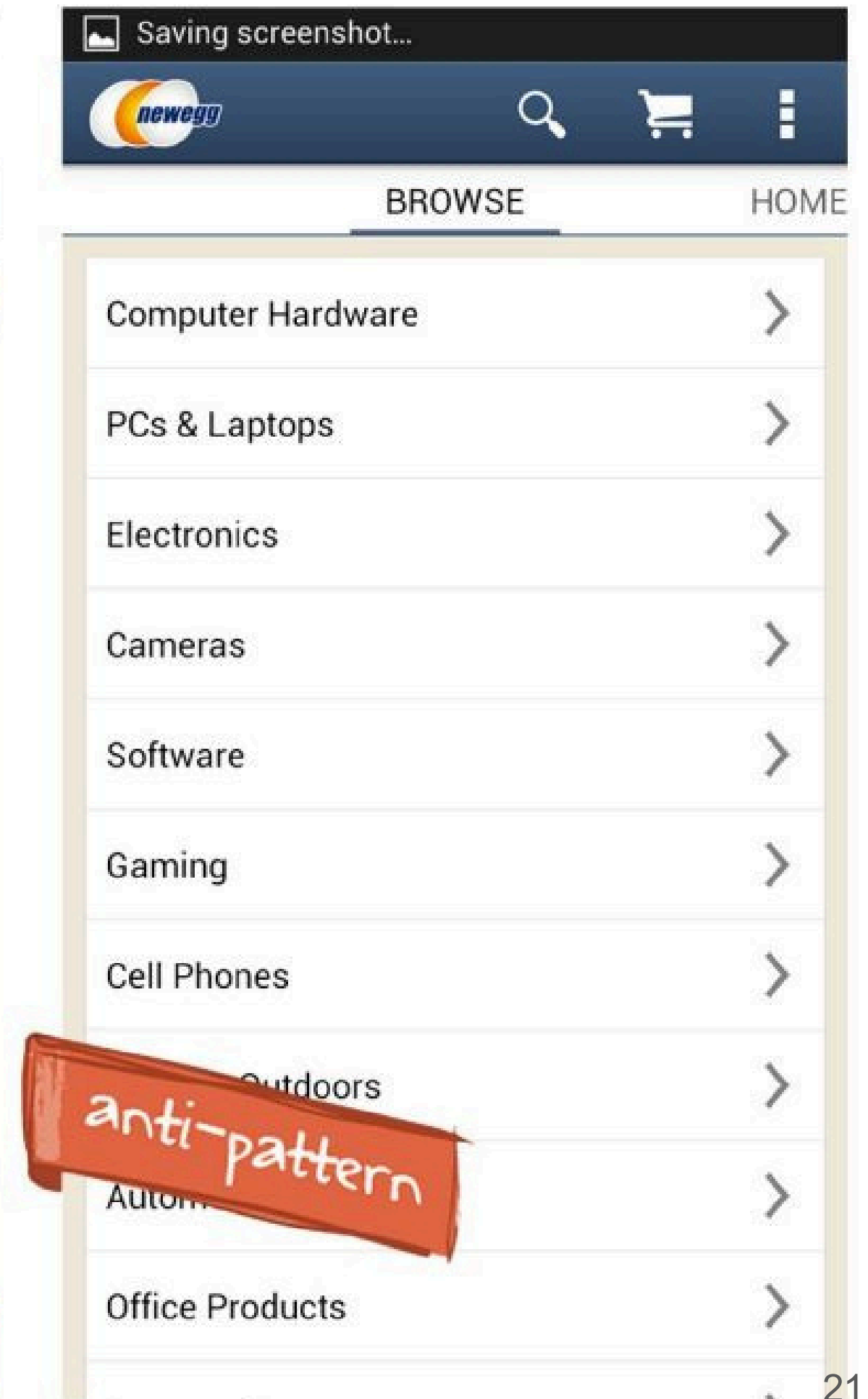
Tab Bar vs Toolbar

It is important to understand the difference between the Tab Bar and Toolbar in iOS. The Tab Bar is for navigating the main categories of the application; the Toolbar presents the tools, or possible actions, for a specific screen.



Designers incorrectly use the Scrolling Tab control

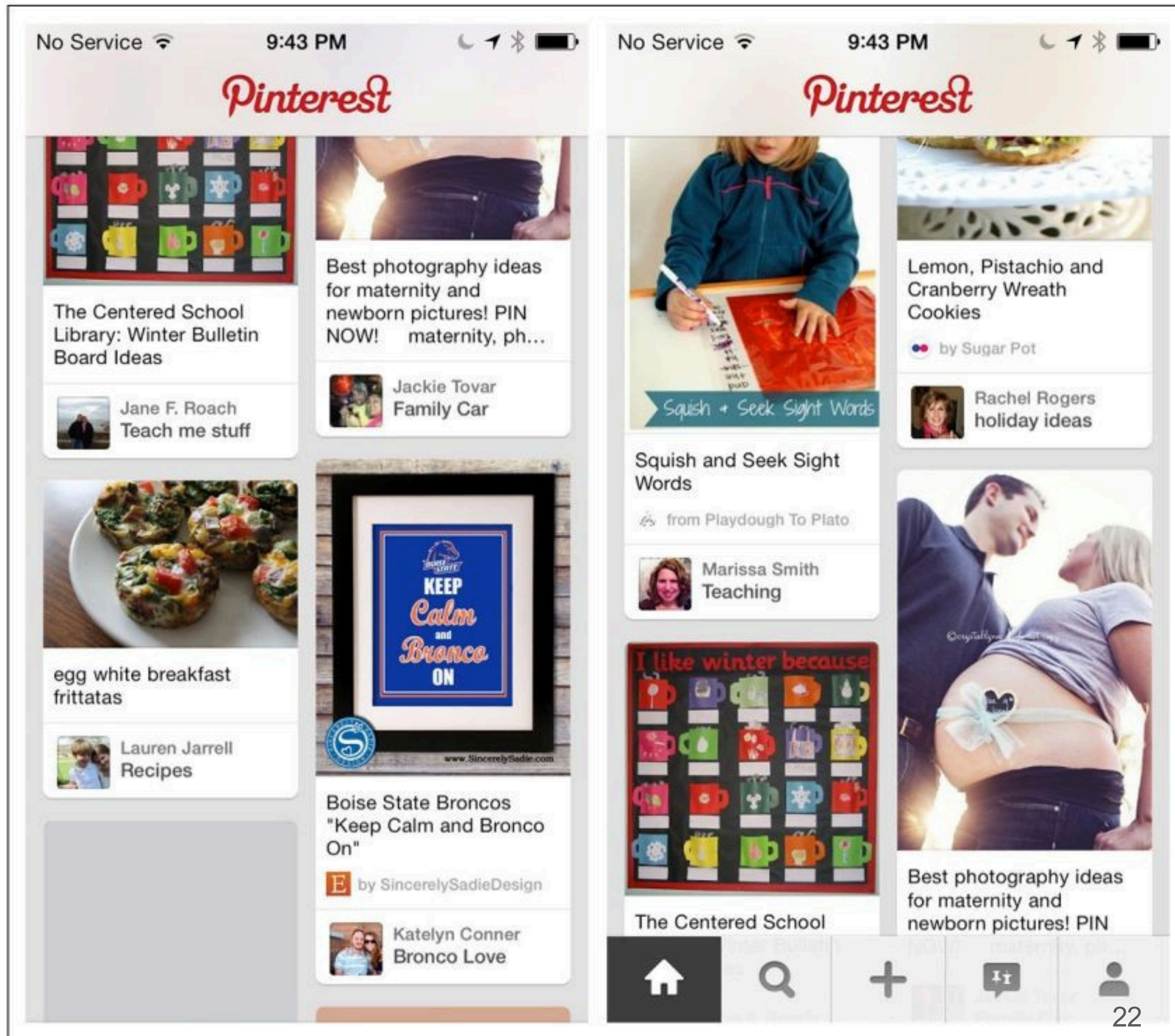
They should have used Side Drawer/Fixed Tab



Emerging Patterns

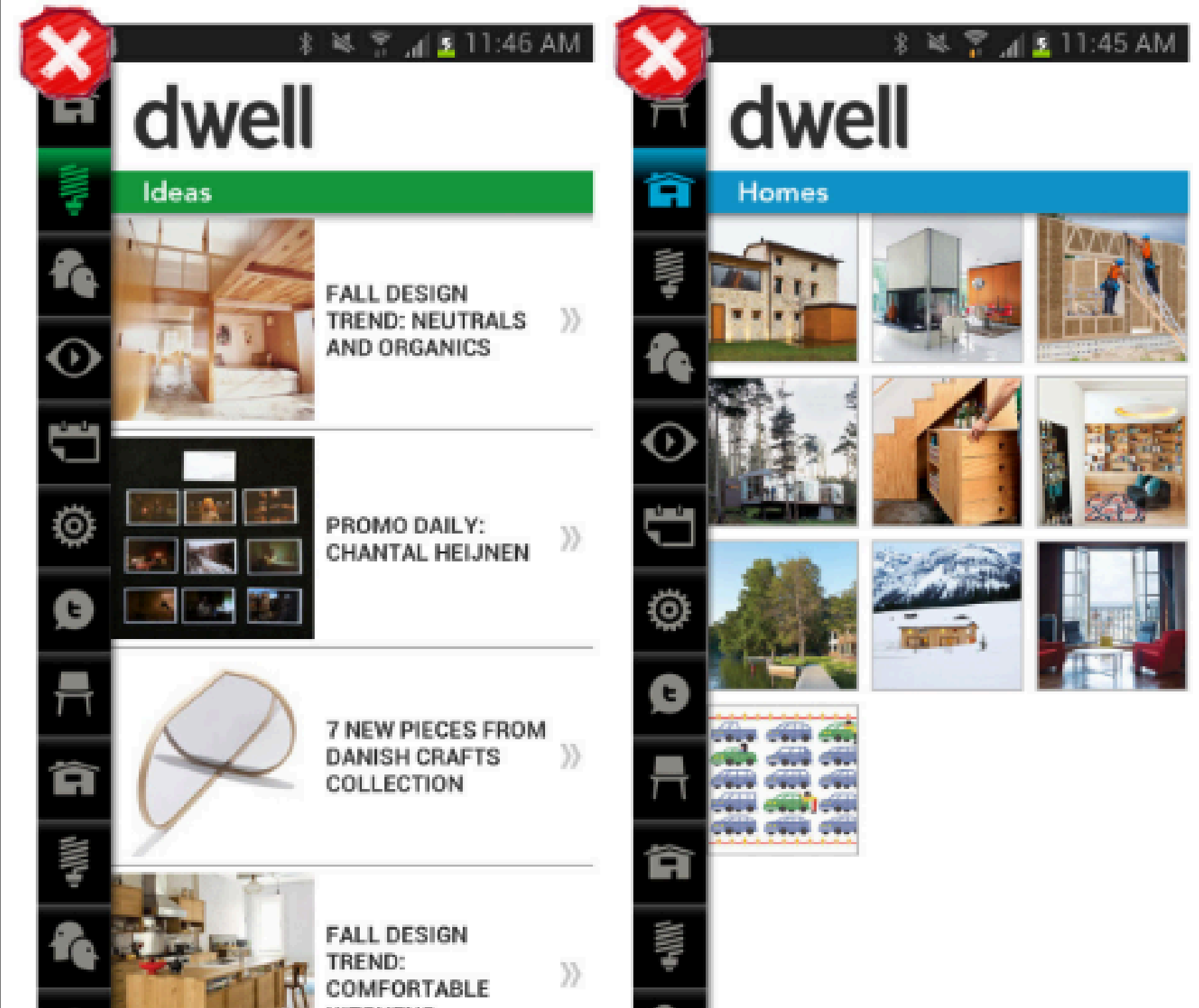
FIGURE 1-33.

Pinterest for iOS:
scrolling down hides
the Toolbar; scrolling
up reveals it ([http://
www.youtube.com/
watch?v=joaaJgvTN28](http://www.youtube.com/watch?v=joaaJgvTN28))



Why Sidebars Are Unsuitable for Mobile Navigation?

- Most people hold their smartphones in portrait mode, and the sidebar takes up a fair amount of horizontal real estate.
- Since the space is so limited, labels will get dropped, which reduces the usability of the app.



Skeuomorphic

The Skeuomorphic pattern is characterized by an interface designed to match its real-world counterpart.

Skeuomorphism is the dominant navigation pattern in game design, as in *Sniper Ghost Warrior 2*, but it can also have its uses in nongame apps, like the music-mixing app *Cross DJ*, the photo app *Hipstamatic*, and the travel app *FlightBoard*.

Examples



T-Mobile 3:41 PM 74%

Airports AUS Departures

Austin-Bergstrom Int'l Airport

FLIGHT	DEST	GATE	DEPARTS
Search Flights			
B6 1094	JFK		1:20 PM
G4 453	LAS		1:25 PM
AA 277	DFW	13	1:33 PM
WN 4569	LAS	12	1:48 PM
WN 689	HOU	8	1:49 PM
US 453	CLT	20	1:50 PM
WN 3197	FLL	11	1:51 PM
WN 2249	PHX	7	1:58 PM
DL 4751	SLC	5	2:11 PM

Flights from 1:40 PM to 1:40 AM CST
Airport Local Time: 3:41 PM CST

T-Mobile 3:41 PM 74%

Airports AUS Departures

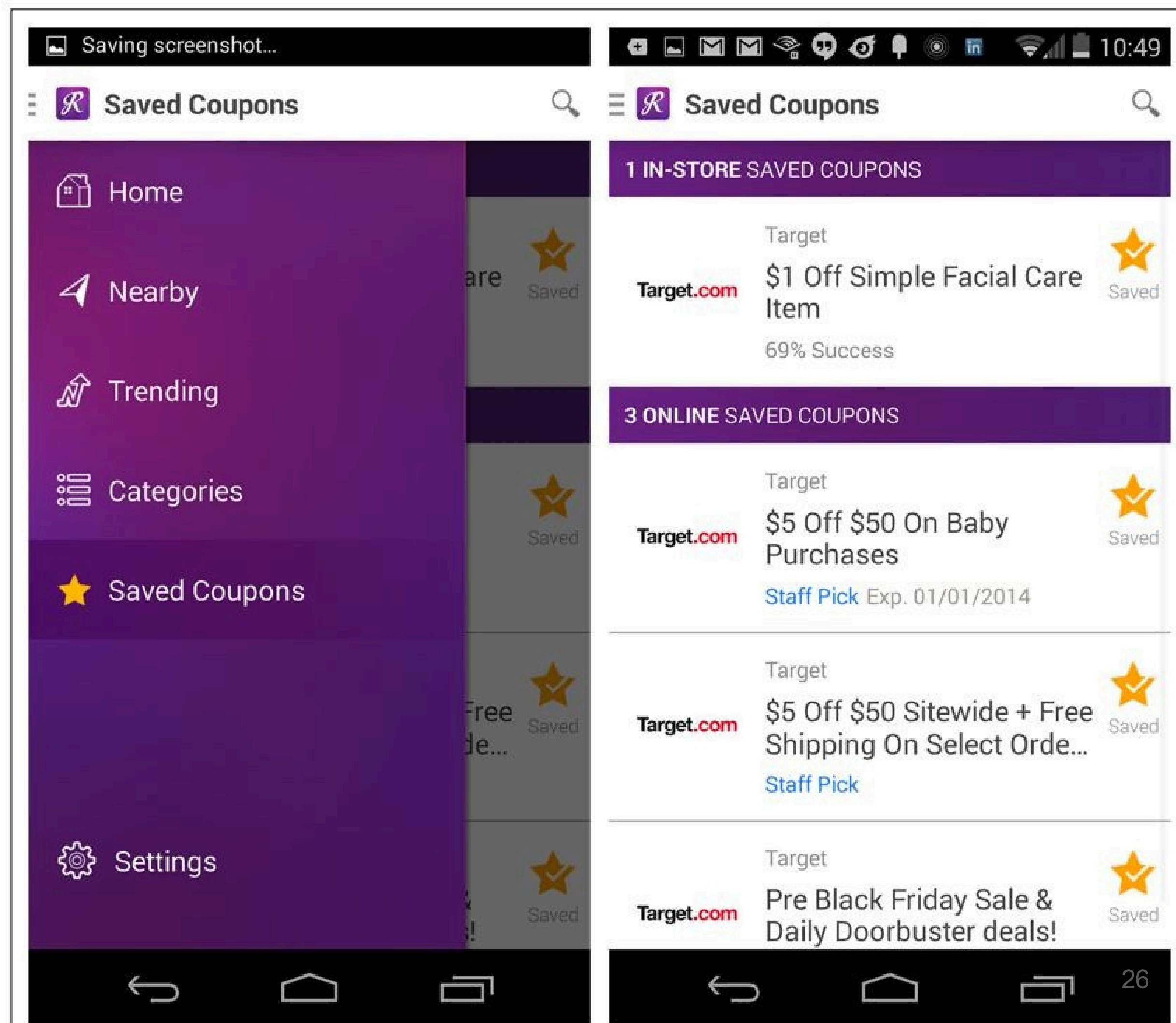
Austin-Bergstrom Int'l Airport

FLIGHT	DEST	GATE	DEPARTS
Search Flights			
B6 1094	JFK		1:20 PM
ALLEGiant AIR (G4) 453 - EN ROUTE			
AUS Austin-Bergstrom Int'l Airport took off on time at 1:25 PM CST			
LAS McCarran Int'l Airport lands 1m early at 2:06 PM PST aircraft Boeing (Douglas) MD-80			
AA 277	DFW	13	1:33 PM
WN 4569	LAS	12	1:48 PM
WN 689	HOU	8	1:49 PM
US 453	CLT	20	1:50 PM

Last updated less than a minute ago

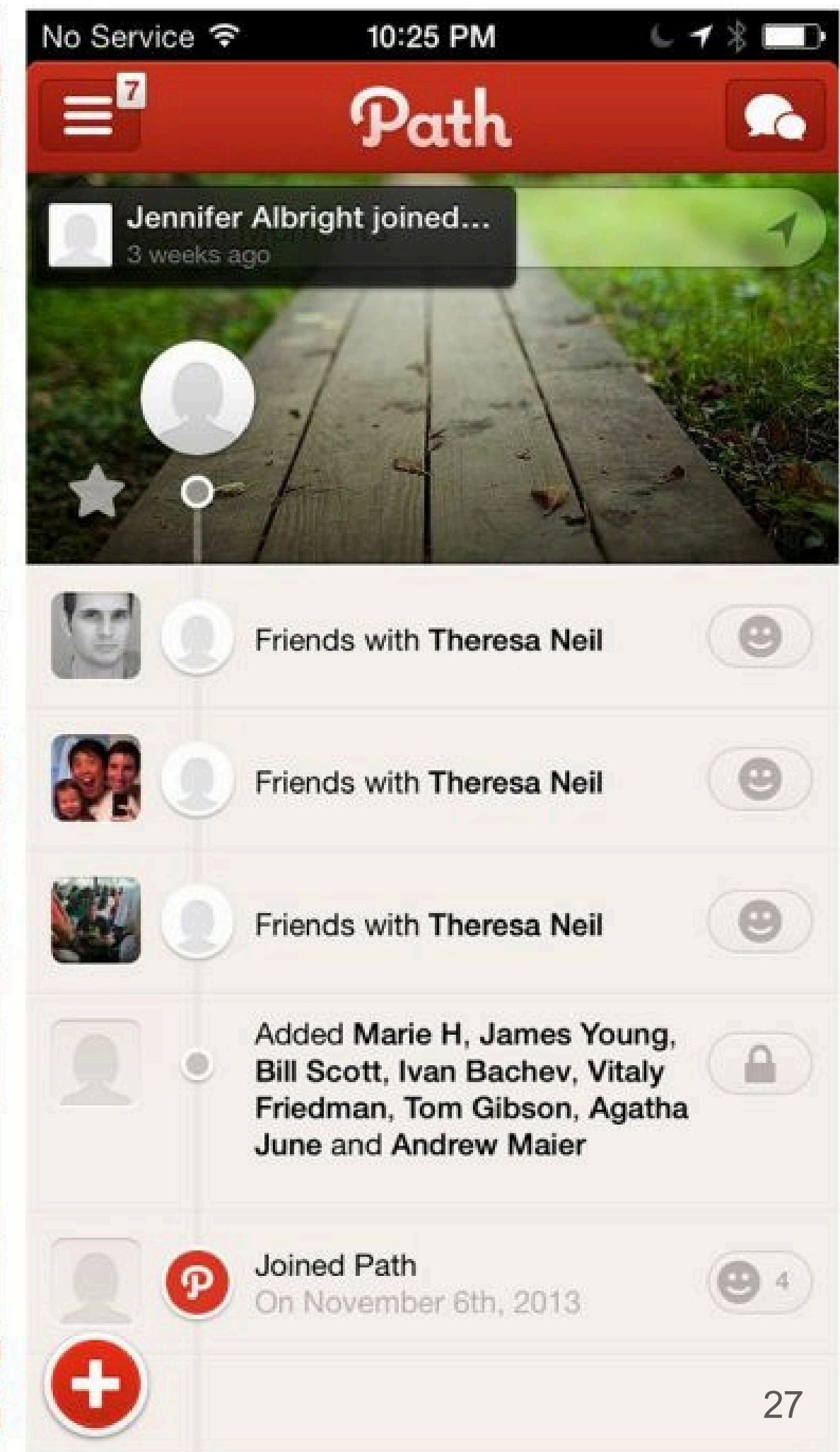
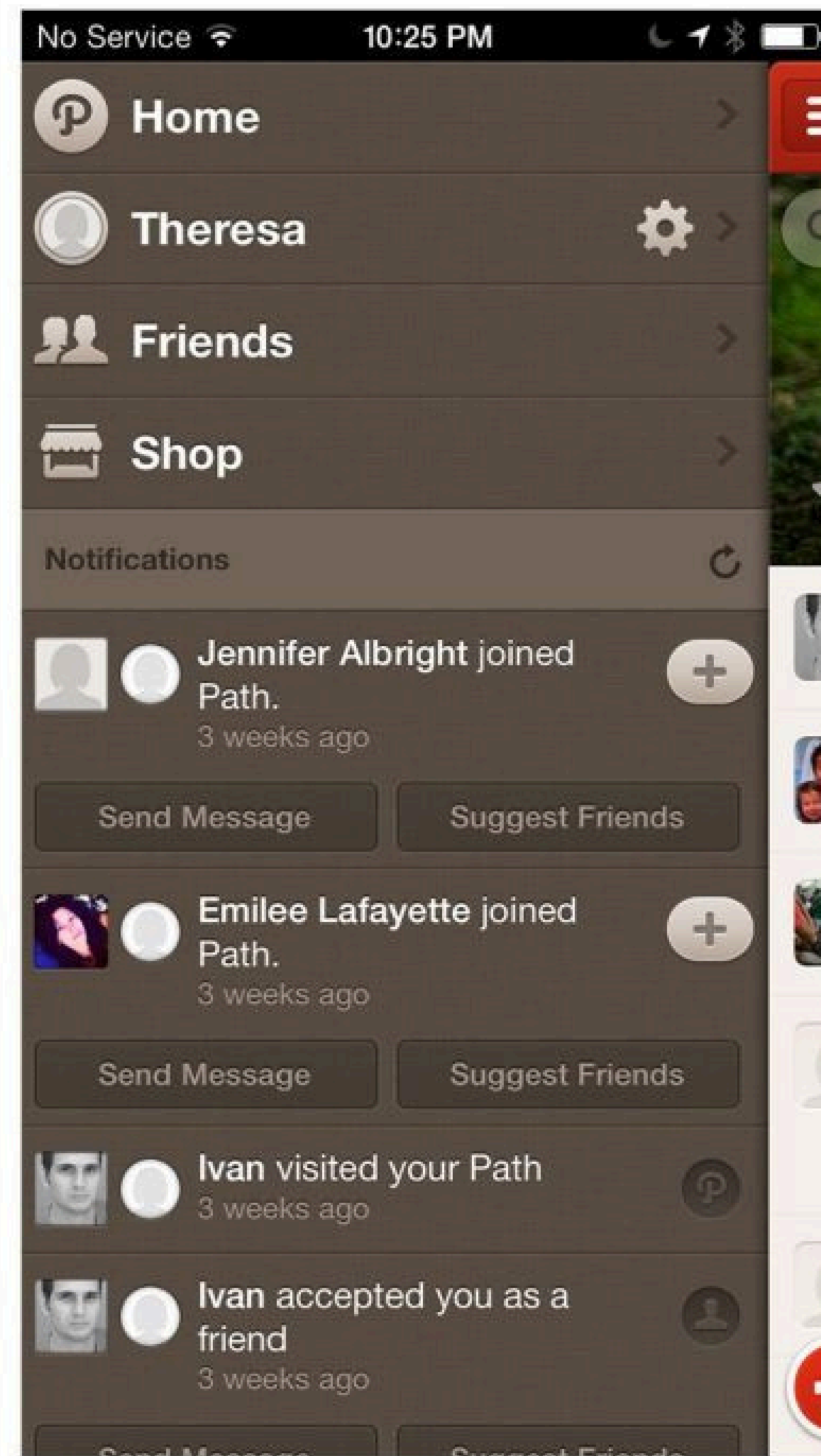
Transient Patterns (Side Drawer)

There are two styles of Side Drawers. The first is an overlay, meaning a swipe or tap gesture will reveal a drawer that partially covers or overlaps the original screen content, as in RetailMeNot. The second style is an inlay, in which a swipe, pan, or tap will open a drawer that pushes the original screen content partially off-canvas, as in Path.



Example

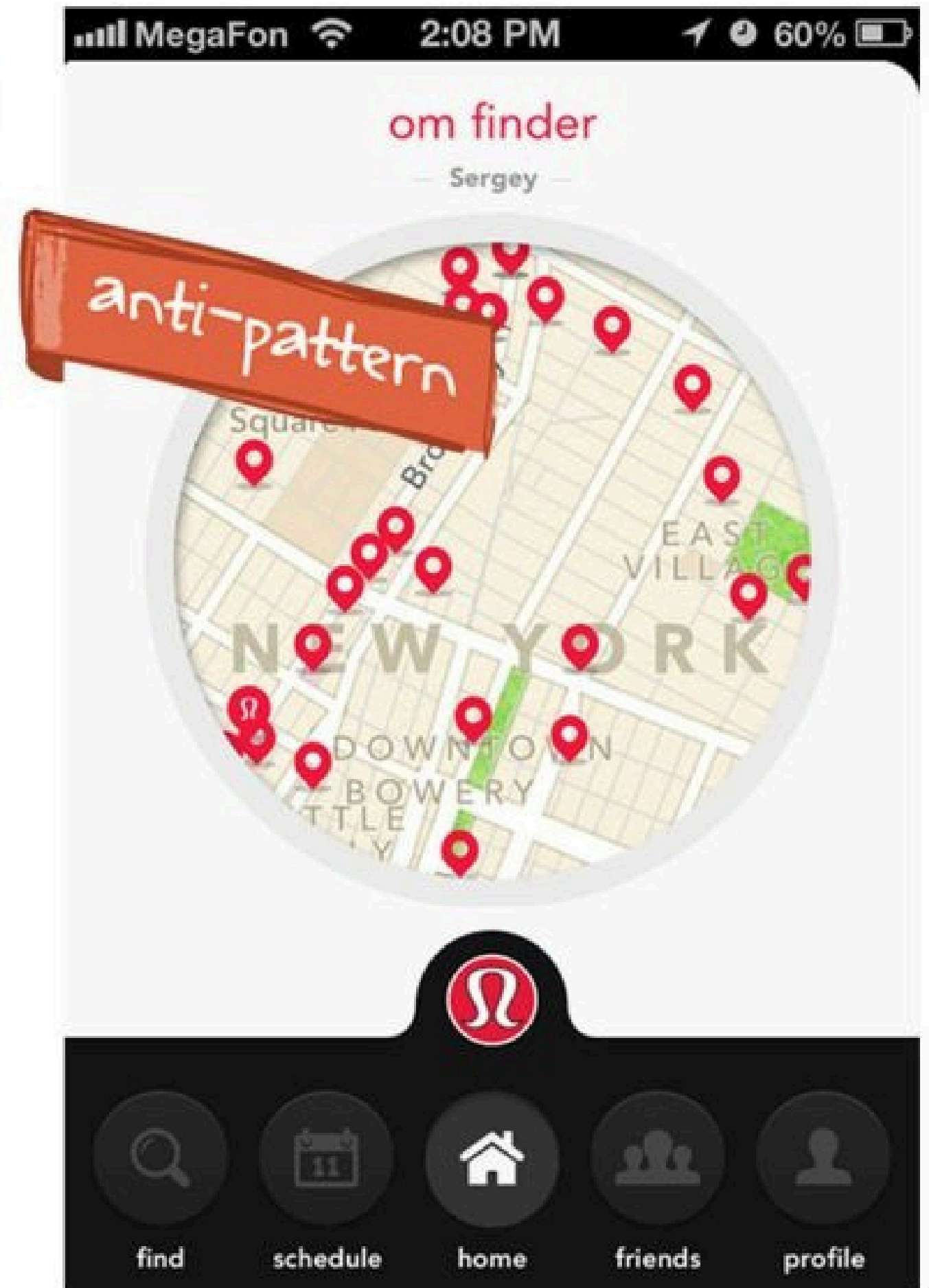
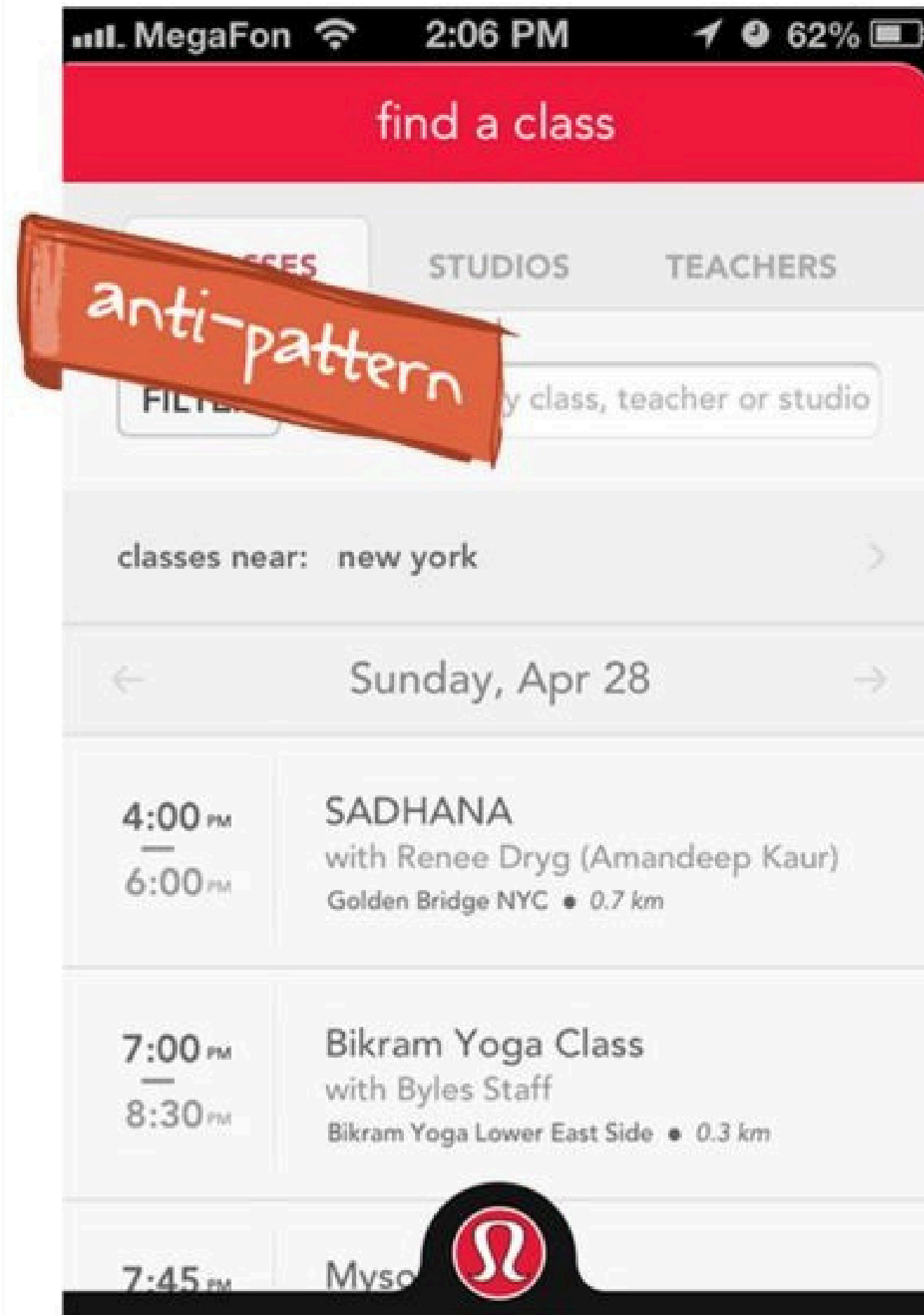
The right side image is inlay or overlay?



Caution

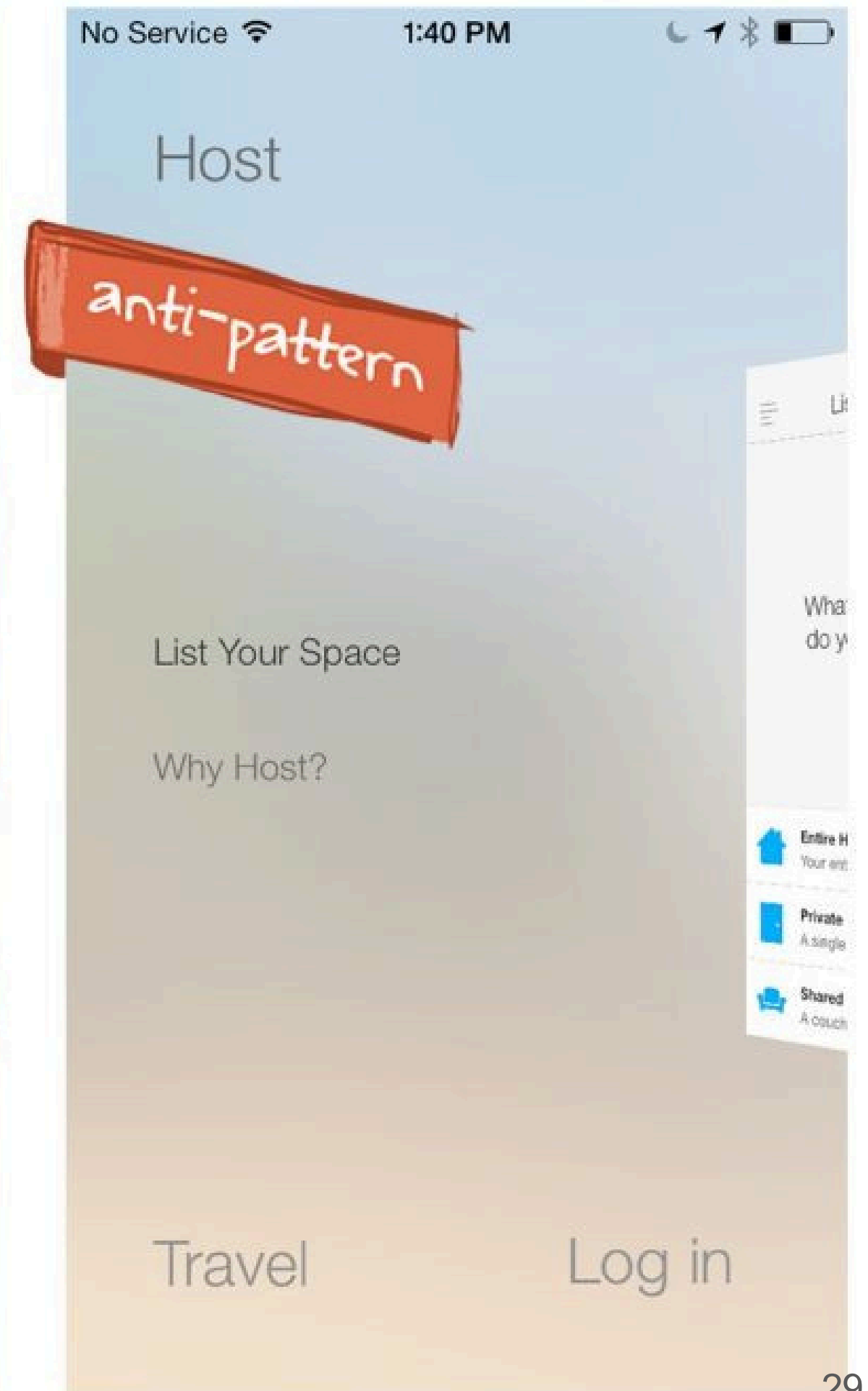
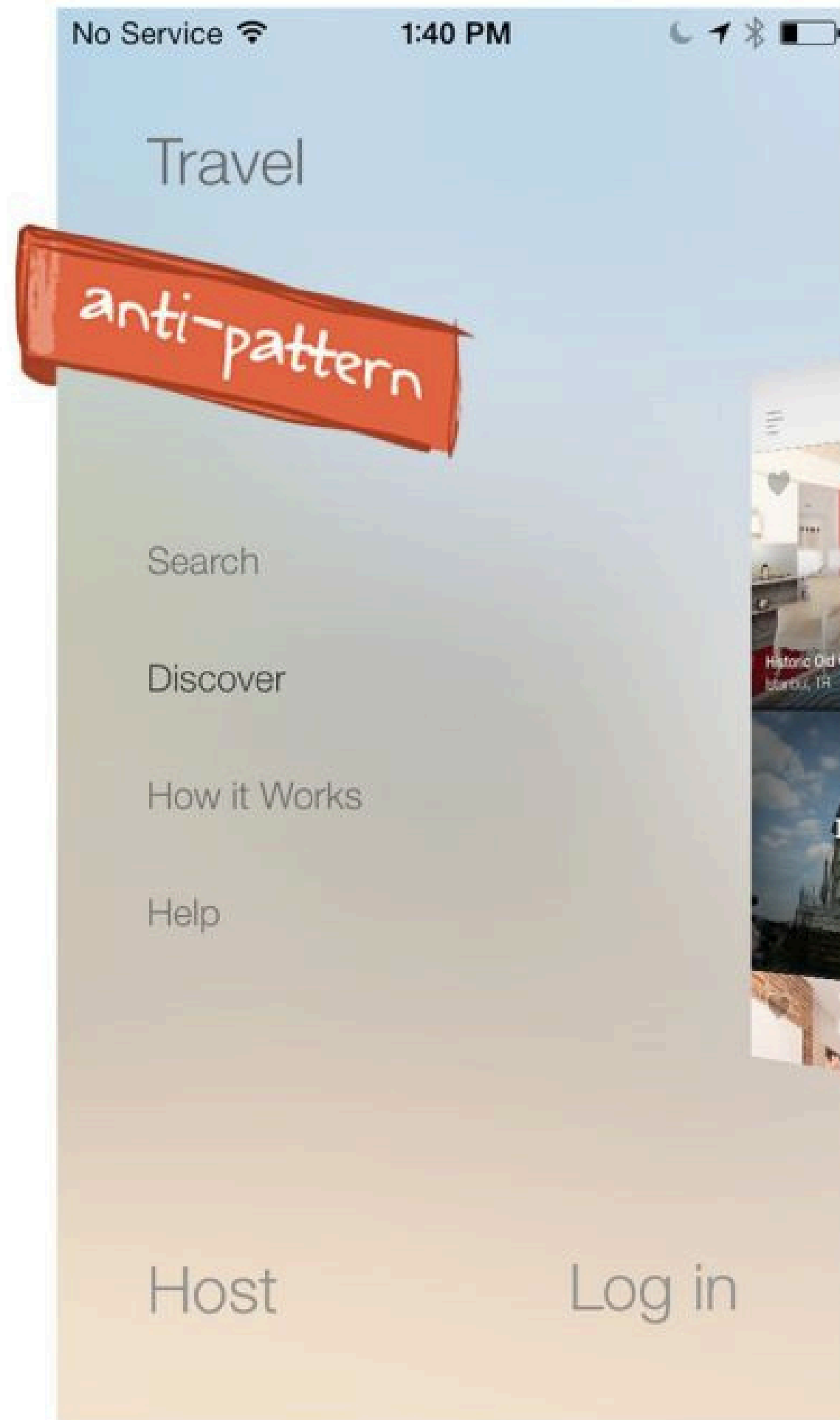
But **don't position the drawer on the bottom**, as in the om finder and Frost apps. This positioning conflicts with the swipe-up gesture that reveals the Control Center in iOS 7.

The Side Drawer can be versatile, but be careful not to overload it with too many features. It **should show the primary navigation options first and foremost**.



Caution

Don't make users scroll to see all their choices. This causes a disconcerting experience for users, because they must spend extra time to distinguish the choices. Also, it can be very difficult for users to scroll without inadvertently tapping an option.



Reference

- **Chapter 1: Navigation**, (Theresa Neil - Mobile Design Pattern Gallery, 2nd Edition_ UI Patterns for Smartphone Apps-O'Reilly Media (2014))